

# Assignment: Instruction Tuning and Direct Preference Optimization on a Small Language Model

Course Staff

## Abstract

This assignment introduces the post-training pipeline used to turn a pretrained language model into a more useful assistant. You will work with a small pretrained model, `Qwen/Qwen2.5-0.5B`, and train it in two stages: supervised fine-tuning (SFT) on instruction-response demonstrations, followed by Direct Preference Optimization (DPO) on pairwise preference data. You will implement the main SFT and DPO training components yourself, run the experiments on Kaggle T4 GPUs, generate responses on a fixed ALPACAEVAL subset, and evaluate model behavior using pairwise judgments from Kaggle model APIs. The goal is not to produce a state-of-the-art assistant, but to understand the mechanics, tradeoffs, and evaluation challenges of modern LLM alignment.

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# 1 Learning Objectives

By the end of this assignment, you should be able to:

|   | Objective                                                                                                         |
|---|-------------------------------------------------------------------------------------------------------------------|
| ✓ | Explain the distinction between pretraining, supervised fine-tuning, RLHF, and DPO.                               |
| ✓ | Train a causal language model with supervised fine-tuning while computing loss only on assistant response tokens. |
| ✓ | Implement the from-scratch DPO loss from pairwise preference data, then compare it with a TRL/LoRA DPO workflow.  |
| ✓ | Train a small language model under realistic GPU and disk constraints.                                            |
| ✓ | Design and run an evaluation pipeline based on fixed prompts, saved generations, and pairwise model judgments.    |
| ✓ | Analyze alignment results critically, including failure cases, noisy judgments, and possible alignment tax.       |

## 2 High-Level Background

### 2.1 Pretraining

Modern language models are first trained with a self-supervised next-token prediction objective. Given a sequence of tokens  $x_1, \dots, x_T$ , the model learns to maximize

$$\sum_{t=1}^T \log p_{\theta}(x_t \mid x_{<t}). \quad (1)$$

This stage teaches the model broad linguistic, factual, and reasoning patterns from large text corpora. However, a pretrained base model is not necessarily a good assistant. It may continue text, imitate undesired formats, fail to follow instructions reliably, or produce unsafe content.

In this assignment, you will *not* pretrain a model. You will start from a pretrained base model, Qwen/Qwen2.5-0.5B, and study post-training.

### 2.2 Supervised Fine-Tuning

Supervised fine-tuning (SFT), also called instruction tuning, trains the model on examples of the form:

`instruction`  $\rightarrow$  `response`.

The model is trained with teacher forcing to assign high probability to the target response. A key implementation detail is that the loss should usually be applied only to response tokens, not to the prompt tokens. If  $x$  is the instruction prompt and  $y$  is the target response, the SFT objective is

$$\mathcal{L}_{\text{SFT}}(\theta) = -\log \pi_{\theta}(y \mid x). \quad (2)$$

SFT teaches a base model to follow an instruction-response format and to imitate high-quality demonstrations.

## 2.3 RLHF

Reinforcement Learning from Human Feedback (RLHF) typically involves three stages:

1. Train an SFT model from demonstrations.
2. Train a reward model from pairwise human preferences.
3. Optimize the language model policy against the reward model, often using PPO with a KL penalty.

RLHF is powerful but complex. It requires reward modeling, online sampling, reinforcement learning machinery, and careful stabilization. Within this broad framework, this assignment focuses on two pieces: supervised fine-tuning (SFT) and policy optimization. For the policy-optimization stage, you will use Direct Preference Optimization (DPO) instead of PPO.

You will *not* implement PPO or a full reward-model-based RLHF pipeline in this assignment. Instead, you will implement DPO, a simpler preference-learning method.

## 2.4 Direct Preference Optimization

Direct Preference Optimization (DPO) is a preference-learning method introduced by Rafailov et al. in *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. It is designed to get many of the benefits of PPO-style policy optimization while avoiding the most complicated parts of a PPO-style reinforcement learning pipeline.

The data for DPO is pairwise. Each training example contains:

instruction  $x$ , chosen response  $y_w$ , rejected response  $y_l$ .

The chosen response is the answer a human or preference dataset judged better. The rejected response is a worse answer to the same instruction. DPO does not ask the model to imitate only the chosen response. Instead, it asks the model to make the chosen response *more likely than the rejected response*, while staying close to a frozen reference model.

### 2.4.1 Why DPO Needs a Reference Model

If we only pushed the policy to prefer chosen responses over rejected responses, the model could drift too far from the SFT model. That can produce strange outputs, repetition, overconfident refusals, or loss of general helpfulness. DPO therefore compares two models:

- the **policy model**  $\pi_\theta$ , which is trainable;
- the **reference model**  $\pi_{\text{ref}}$ , which is frozen and usually initialized from the same SFT checkpoint.

The reference model acts like an anchor. DPO rewards the policy for improving the chosen-vs-rejected preference *relative to what the SFT model already believed*. This is the part that replaces the explicit KL penalty used in many RLHF pipelines.

### 2.4.2 The Loss, Intuitively

For each pair, compute four response log probabilities:

| Quantity                         | Meaning                                            |
|----------------------------------|----------------------------------------------------|
| $\log \pi_\theta(y_w   x)$       | policy log probability of the chosen response      |
| $\log \pi_\theta(y_l   x)$       | policy log probability of the rejected response    |
| $\log \pi_{\text{ref}}(y_w   x)$ | reference log probability of the chosen response   |
| $\log \pi_{\text{ref}}(y_l   x)$ | reference log probability of the rejected response |

Then compare two preference gaps:

$$\Delta_\theta = \log \pi_\theta(y_w | x) - \log \pi_\theta(y_l | x), \quad (3)$$

$$\Delta_{\text{ref}} = \log \pi_{\text{ref}}(y_w | x) - \log \pi_{\text{ref}}(y_l | x). \quad (4)$$

$\Delta_\theta$  says how much the trainable model prefers the chosen response over the rejected response.  $\Delta_{\text{ref}}$  says how much the original SFT model already preferred the chosen response. DPO trains the policy so that  $\Delta_\theta$  becomes larger than  $\Delta_{\text{ref}}$ .

The per-example DPO loss is:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma(\beta [(\log \pi_\theta(y_w | x) - \log \pi_\theta(y_l | x)) - (\log \pi_{\text{ref}}(y_w | x) - \log \pi_{\text{ref}}(y_l | x))]). \quad (5)$$

Equivalently:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left( \beta \left[ \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right] \right). \quad (6)$$

The hyperparameter  $\beta$  controls how strongly the preference signal is applied. A larger  $\beta$  makes the optimization more aggressive. A smaller  $\beta$  keeps the policy closer to the reference model. In this assignment,  $\beta = 0.1$  is the recommended starting point.

#### DPO in One Sentence

DPO teaches the model to increase the probability gap between preferred and rejected answers, measured relative to a frozen SFT reference model, without training a separate reward model or running online reinforcement learning.

### 2.4.3 Step-by-Step DPO Training

A single DPO update looks like this:

1. Format the instruction  $x$  exactly the same way for both the chosen and rejected response.
2. Tokenize the prompt-response sequence for  $y_w$  and for  $y_l$ .
3. Mask prompt tokens so that log probabilities are summed only over response tokens.
4. Run the trainable policy model on the chosen and rejected sequences.
5. Run the frozen reference model on the same chosen and rejected sequences with gradients disabled.
6. Sum token log probabilities over the response span for each sequence.
7. Compute the policy preference gap  $\Delta_\theta$  and the reference preference gap  $\Delta_{\text{ref}}$ .
8. Apply the DPO logistic loss  $-\log \sigma(\beta(\Delta_\theta - \Delta_{\text{ref}}))$ .

9. Backpropagate only through the policy model.
10. Repeat with gradient accumulation, validation, and checkpointing.

The important implementation idea is that DPO is still ordinary supervised optimization over saved data. There is no environment, no online sampling loop, and no separate reward-model training stage.

### 3 What You Will Build

You will build a complete small-model alignment pipeline with three preference-optimization tracks:

1. Create a reproducible GitHub repository and Kaggle workflow.
2. Generate baseline responses from Qwen/Qwen2.5-0.5B on fixed evaluation subsets.
3. Train one SFT model on 30,000 instruction-response examples.
4. Run from-scratch DPO on Anthropic Helpful-Harmless-style preference data, then test it on the synthetic helpful/harmless evaluation set provided with the assignment.
5. Run from-scratch DPO on UltraFeedback-style preference data, then test it on the fixed ALPACAEVAL subset using the same seed across all models.
6. Run a TRL DPOTrainer + LoRA experiment on the provided persona preference subset in `data/persona_lora/ramsay_style_preferences.json`.
7. Generate outputs, build pairwise comparison files, and evaluate with model-based judgments.
8. Submit a complete written report comparing the SFT model, both from-scratch DPO models, and the LoRA-DPO model.

#### Library Boundary

You may use [Hugging Face Transformers](#) to load models and tokenizers. You may use PyTorch utilities such as `DataLoader`, optimizers, schedulers, and AMP. For the SFT implementation and the two from-scratch DPO runs, you may **not** use Hugging Face `Trainer`, `TRL DPOTrainer`, or a prebuilt DPO training loop. The only exception is the LoRA persona task, where you **should** use TRL's `DPOTrainer` with PEFT/LoRA.

#### Guide Configurations, Not Sacred Hyperparameters

The example sizes and hyperparameters in this manual are the reference recipe used to make the assignment feasible on Kaggle T4 GPUs. They are not guaranteed to be optimal. Students may try other learning rates, optimizers, batch sizes, data mixtures, LoRA ranks, or training lengths if they document the changes and compare results fairly.

### 4 Private Evaluation Interface

The course staff may use a private grader to check important implementation details, such as response-token masking, log-probability computation, and DPO loss correctness. Students will not

receive direct access to this grader.

Your repository should still be organized clearly enough that staff can inspect and run your implementation. In particular, make sure your training code is easy to locate, your notebooks use reproducible paths, and your report explains any deviations from the guide configuration.

## 5 Compute Environment

### 5.1 Kaggle Accelerator

Use a Kaggle notebook with the GPU T4 x2 accelerator. You should verify that two GPUs are visible:

```
nvidia-smi
```

A typical output should show two Tesla T4 GPUs with roughly 15 GiB of memory each.

### 5.2 Dependency Guidance

Kaggle already includes a CUDA-compatible PyTorch build. Do not reinstall PyTorch unless you have a specific reason. A safe install pattern is:

```
python -m pip uninstall -y -q torchao || true
python -m pip install -q "transformers>=4.43.0,<5" "datasets>=2.18,<5" \
    huggingface_hub accelerate tqdm pyyaml sentencepiece safetensors
python -m pip install -q -e . --no-deps
```

This setup is useful because Kaggle’s preinstalled PyTorch is already matched to the notebook GPU runtime. Reinstalling PyTorch can waste time, pull the wrong CUDA build, or break GPU access. The optional `exttttorchao` uninstall avoids a known compatibility issue where older Kaggle `exttttorchao` builds can conflict with PEFT/LoRA loading. Installing the starter repo with `exttt-e . --no-deps` makes local imports work without letting `extttpip` replace the carefully managed Kaggle environment.

Your repository should include a minimal `requirements.txt` and `pyproject.toml`. The requirements should not pin `torch`. A suitable `requirements.txt` is:

```
transformers>=4.43.0,<5
datasets>=2.18,<5
huggingface_hub
accelerate
tqdm
pyyaml
sentencepiece
safetensors
```

### 5.3 Starter Repository

Students will receive restricted access to a clean starter repository branch, not the staff/reference solution codebase. This branch exists so students can build on top of a tested environment setup without receiving completed SFT or DPO code.

The starter branch should contain only the essentials:

► `requirements.txt`;

- ▶ `pyproject.toml`;
- ▶ fixed data files under `data/`;
- ▶ fixed evaluation files under `eval_sets/`;
- ▶ the assignment PDF.

A minimal starter layout is:

```
small-model-alignment-starter/  
  requirements.txt  
  pyproject.toml  
  data/  
    persona_lora/  
      ramsay_style_preferences.json  
      README.md  
  eval_sets/  
    hh_rlhf_eval_300_seed42.json  
  docs/  
    assignment_manual.pdf
```

You may also provide a separate example codebase with general package and run names. If you do so, it should illustrate the workflow and file organization without exposing staff solution code or course-specific private implementation details.

#### Starter Repo Sanity Check

Before sharing the starter branch, clone it into a fresh environment and verify that the setup commands run successfully. Students should not hit avoidable errors from missing files, stale package names, or broken paths before they have written any training code.

## 5.4 Kaggle Workflow

A practical workflow is:

1. Fork or copy the starter repository, then develop and push your code to GitHub.
2. Clone the repository in Kaggle.
3. Run training in one notebook.
4. Save the Kaggle notebook version so that outputs are preserved.
5. Attach the training notebook outputs as an input dataset to a separate generation notebook.
6. Attach generation outputs as an input dataset to a separate evaluation notebook.

This separation avoids rerunning expensive training when debugging evaluation.

## 5.5 Saving Work While Kaggle Jobs Run

Kaggle notebook sessions are temporary. Anything important must be written under `/kaggle/working` and then preserved by saving a notebook version or creating an output dataset. The

safest pattern is to make every long-running stage write durable artifacts as it goes, rather than only at the very end.

For training notebooks:

1. Write checkpoints, metrics, logs, and plots under a stable run directory such as `/kaggle/working/<repo_name>/outputs/<run_name>/`.
2. Save at least **best** and **final** checkpoints. Avoid saving too many full intermediate checkpoints because Kaggle disk is limited.
3. Write a small metrics file after every evaluation step, for example `metrics.jsonl`, so partial progress survives if the notebook disconnects.
4. When a training stage completes, use **Save Version** in Kaggle. Saved versions preserve the notebook outputs and can be attached to later notebooks as input datasets.
5. For extra safety, create a zip archive in `/kaggle/working`, for example `sft_run_outputs.zip` or `dpo_run_outputs.zip`. Zips are easier to attach, download, and audit.

For generation and evaluation notebooks:

1. Write generation JSON files immediately after each model finishes.
2. Write pairwise JSONL files before judging, not inside the judging loop.
3. During model-based judging, append or rewrite judgment files after every batch so the run can resume from existing outputs.
4. Keep failure logs, retry counts, and summary files in the same output directory.

A useful mental model is: every notebook should be restartable. If Kaggle disconnects halfway through, you should be able to attach the previous output dataset, skip completed work, and continue from the last saved artifact.

## 6 Model and Data

### 6.1 Model

Use `Qwen/Qwen2.5-0.5B` as the base model. This model is small enough to train on Kaggle T4 GPUs, while still being large enough to show meaningful differences between base, SFT, and DPO behavior.

### 6.2 SFT Data

Use an instruction-response dataset such as UltraChat-style data. Each example should be converted into a single-turn instruction and response.

Recommended run size:

| Split      | Examples | Purpose          |
|------------|----------|------------------|
| Train      | 30,000   | SFT optimization |
| Validation | 1,000    | validation loss  |

### 6.3 Preference Data

The assignment uses three preference datasets, each with examples containing a prompt, a chosen response, and a rejected response:

| Track                    | Data                                                                                                                                                   | Recommended train size |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>HH-RLHF DPO</b>       | Helpful/harmless preference pairs. Use this for the safety-oriented from-scratch DPO run.                                                              | 15,000                 |
| <b>UltraFeedback DPO</b> | General instruction-following preference pairs. Use this for the broad helpfulness from-scratch DPO run.                                               | 15,000                 |
| <b>LoRA persona DPO</b>  | Provided small preference subset in <code>data/persona_lora/ramsay_style_preferences.json</code> . Use this with <code>TRL DPOTrainer</code> and LoRA. | provided subset        |

Use 1,000 validation pairs for each from-scratch DPO run when available. You should filter out multi-turn examples if your preprocessing assumes single-turn prompts.

### 6.4 Evaluation Prompts

Use fixed evaluation sets so that models are compared on the same prompts:

| Training track           | Evaluation set                                                                                                  |
|--------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>HH-RLHF DPO</b>       | The synthetic helpful/harmless evaluation set provided with the assignment.                                     |
| <b>UltraFeedback DPO</b> | The fixed ALPACAEVAL subset generated with seed 42. The default is 300 prompts.                                 |
| <b>LoRA persona DPO</b>  | A small fixed style-evaluation prompt set covering cooking, coding, general help, and safety-sensitive prompts. |

Save the exact subsets used for all comparisons. To reproduce the default ALPACAEVAL evaluation subset, use 300 prompts with seed 42.

## 7 Task 1: Repository and Kaggle Setup

Create a GitHub repository and a Kaggle notebook workflow. Your setup should:

1. Clone the starter repository, or your fork of it, in Kaggle.
2. Install dependencies without reinstalling PyTorch.
3. Confirm that two T4 GPUs are available.
4. Print package versions for PyTorch, Transformers, and Datasets.

## 8 Task 2: Baseline Generation

Create a fixed ALPACAEVAL subset and generate baseline responses from Qwen/Qwen2.5-0.5B.

Your generation script should save a JSON file with entries containing at least:

```
{
  "id": "...",
  "instruction": "...",
  "output": "...",
  "generator": "qwen2.5-0.5b-base",
  "dataset": "alpaca_eval"
}
```

Use greedy decoding for the main comparison unless you justify another choice. Small models sometimes fall into repetitive loops; if this happens, you may use deterministic anti-repetition controls such as a modest repetition penalty or a no-repeat n-gram constraint. Document any decoding changes so SFT and DPO are compared fairly.

## 9 Task 3: Supervised Fine-Tuning

### 9.1 Implementation Requirements

Implement your own SFT training loop. Your implementation should include:

1. Prompt formatting for instruction-response examples.
2. Response-only loss masking.
3. Padding and attention masks.
4. Gradient accumulation.
5. Mixed precision using fp16 autocast on T4 GPUs.
6. FP32 trainable weights when using gradient scaling.
7. Gradient clipping.
8. Validation loss.
9. Saving **best** and **final** checkpoints.

Do not save frequent full intermediate checkpoints on Kaggle unless you have enough disk space. Saving only **best** and **final** is sufficient.

## 9.2 Suggested Hyperparameters

| Hyperparameter        | Value              |
|-----------------------|--------------------|
| Model                 | Qwen/Qwen2.5-0.5B  |
| Context length        | 512                |
| Train examples        | 30,000             |
| Validation examples   | 1,000              |
| Per-device batch size | 2                  |
| Gradient accumulation | 16                 |
| Effective batch size  | 32                 |
| Learning rate         | $2 \times 10^{-5}$ |
| Warmup                | 3%                 |
| Schedule              | cosine decay       |
| Weight decay          | 0.1                |
| Gradient clipping     | 1.0                |
| Epochs                | 1                  |

## 10 Task 4: Two From-Scratch DPO Runs

### 10.1 Implementation Requirements

Implement DPO yourself and use the same core implementation for both preference datasets. Your implementation should include:

1. Loading the SFT model as the trainable policy.
2. Loading a frozen copy of the SFT model as the reference model.
3. Computing response-only log probabilities for chosen and rejected responses.
4. Computing the DPO loss.
5. Tracking validation DPO loss, reward accuracy, and reward margin.
6. Saving the best checkpoint according to validation reward accuracy.

### Kaggle Memory Tip

A memory-friendly setup on Kaggle T4x2 is to place the policy on `cuda:0` and the frozen reference on `cuda:1`. The trainable policy should remain in fp32 when using gradient scaling; the frozen reference may be loaded in fp16.

## 10.2 Required DPO Runs

After SFT, run two separate DPO trainings from the same SFT checkpoint:

1. **HH-RLHF DPO**: train on 15,000 helpful/harmless preference pairs and evaluate on the synthetic helpful/harmless set included with the assignment.
2. **UltraFeedback DPO**: train on 15,000 UltraFeedback-style preference pairs and evaluate on the fixed ALPACAEVAL subset using the same seed used for the base and SFT generations.

These are two different alignment experiments. HH-RLHF emphasizes safety and refusal behavior. UltraFeedback is broader and usually emphasizes general helpfulness, instruction-following, and response quality.

## 10.3 Reference Training Recipe

The guide configuration is:

1. Train one SFT checkpoint from Qwen/Qwen2.5-0.5B on 30,000 examples.
2. Start each DPO run from the SFT `best` checkpoint.
3. Load a second frozen copy of the SFT `best` checkpoint as the reference model.
4. Use the same prompt template in DPO that was used in SFT. For the main Qwen run, this means `plain`.
5. Use response-only log probabilities: prompt tokens and padding tokens must not contribute to the chosen or rejected sequence score.
6. Use  $\beta = 0.1$  and a small fixed learning rate,  $1 \times 10^{-6}$ .
7. Use gradient accumulation so the effective batch size is 32 even though each device sees only one pair at a time.
8. Save the checkpoint with the best validation reward accuracy, plus a final checkpoint.

The reference DPO run uses a small fixed learning rate. You may try schedules such as warmup or cosine decay, but if you do, report the change clearly.

## 10.4 Suggested Hyperparameters

| Hyperparameter        | Value                                           |
|-----------------------|-------------------------------------------------|
| Initial policy        | SFT <b>best</b> checkpoint                      |
| Reference model       | frozen SFT <b>best</b> checkpoint               |
| Prompt format         | same as SFT; <b>plain</b> for the main Qwen run |
| Train pairs           | 15,000 per DPO run                              |
| Validation pairs      | 1,000 per DPO run                               |
| Context length        | 512                                             |
| Maximum prompt length | 256                                             |
| Per-device batch size | 1                                               |
| Gradient accumulation | 32                                              |
| Effective batch size  | 32                                              |
| Learning rate         | $1 \times 10^{-6}$ , fixed                      |
| Learning-rate decay   | none by default for DPO                         |
| $\beta$               | 0.1                                             |
| Optimizer             | RMSprop or AdamW                                |
| Gradient clipping     | 1.0                                             |
| Epochs                | 1                                               |
| Checkpoint selection  | best validation reward accuracy                 |

## 10.5 Metrics

For each validation pair, define the implicit rewards:

$$r_w = \beta (\log \pi_\theta(y_w | x) - \log \pi_{\text{ref}}(y_w | x)), \quad (7)$$

$$r_l = \beta (\log \pi_\theta(y_l | x) - \log \pi_{\text{ref}}(y_l | x)). \quad (8)$$

Reward accuracy is the fraction of validation examples where

$$r_w > r_l. \quad (9)$$

Reward margin is the mean value of

$$r_w - r_l. \quad (10)$$

A reward accuracy near 0.5 with margin near 0 suggests little preference signal. Higher reward accuracy with a positive margin suggests the policy is learning to prefer chosen responses.

## 11 Task 5: Idiot Sandwich — TRL DPOTrainer With LoRA Persona Tuning

This task uses TRL’s DPOTrainer and PEFT/LoRA to train a small instruction model on the provided persona preference subset. Unlike Task 4, this task **does** allow TRL and PEFT. The working title is *Idiot Sandwich*: a small style-steering experiment inspired by the famous cooking meme, while still requiring the model to remain useful and safe.



Use:

HuggingFaceTB/SmolLM2-360M-Instruct

Train on:

data/persona\_lora/ramsay\_style\_preferences.json

Each row contains `prompt`, `chosen`, and `rejected`. Format it into the conversational structure expected by TRL.

Do not treat the LoRA setup as a copy-paste exercise. Read the [PEFT documentation](#) and the [TRL DPOTrainer documentation](#), then choose an appropriate LoRA configuration for the model. Your report should briefly justify the LoRA rank, target modules, learning rate, batch size, and number of epochs you used.

Evaluate whether the LoRA-DPO model became more stylistically faithful without becoming less useful or less safe.

## 12 Task 6: Generate Evaluation Outputs

Before launching full-subset generation, run a small sanity check on 2–5 prompts. Inspect the responses manually to confirm that the checkpoint path, tokenizer, prompt format, EOS behavior, and generation length are reasonable.

Generate outputs from at least:

1. the SFT model;
2. the DPO model being compared against it.

For the UltraFeedback DPO run, use the fixed ALPACAEVAL subset generated with 300 prompts and seed 42. For the HH-RLHF DPO run, use the provided synthetic helpful/harmless evaluation set.

Build pairwise comparison files for **DPO versus SFT**. Randomize response order in each pairwise file and preserve the mapping from response position to model identity. You may generate base-model outputs for your own analysis, but the required comparison is DPO vs SFT.

#### Generation Coherence Tip

If outputs from the small models become repetitive, you may try deterministic anti-repetition settings such as a repetition penalty or no-repeat n-gram constraint. Use the same decoding settings for SFT and DPO in a comparison so the judge is evaluating the models, not different decoding policies.

## 13 Task 7: Evaluation With Kaggle Models

Use Kaggle model access through `kaggle_benchmarks` or an equivalent Kaggle-supported model interface. A suitable judge model is:

`google/gemini-3.6-flash`

If that model is unavailable in your Kaggle environment, you may use another supported Kaggle model-proxy judge. Record the judge model you used in your report.

### 13.1 Kaggle Benchmark Setup

For model-based evaluation, create or use a Kaggle notebook that is configured as a **utility benchmark**. This is important because Kaggle’s benchmark/model-proxy environment allows free model calls to supported judge models such as `google/gemini-3.6-flash`. A practical setup is:

1. Save generation outputs from the training/generation notebook as a Kaggle output dataset.
2. Create a separate evaluation notebook and attach the generation-output dataset as an input.
3. Enable the Kaggle benchmark/model access needed by `kaggle_benchmarks`.
4. Run judging from saved pairwise JSONL files only. Do not regenerate model outputs inside the judge notebook.
5. Write judgment files back to `/kaggle/working` and save the evaluation notebook version when judging completes.

The evaluation notebook should be treated as a resumable batch job. It should not assume that all judge calls succeed on the first attempt.

## 13.2 Evaluator Reliability Requirements

Your evaluator should:

1. Read saved pairwise JSONL files.
2. Judge in small batches, such as 5 pairs per request.
3. Checkpoint judgments after each batch.
4. Sleep at least 5 seconds between judge request attempts to avoid overloading the model proxy or receiving request-denied errors.
5. Retry failed batches multiple times, for example up to 3 attempts per batch.
6. Log unresolved failures to a separate JSONL failure file.
7. Resume from existing judgments when possible so reruns do not spend calls on completed pairs.
8. Summarize win, loss, tie, unknown, and failure rates.

A representative command is:

```
python scripts/judge_pairwise_kaggle.py \  
  --pairs_path outputs/eval/dpo_vs_sft_pairs.jsonl \  
  --output_path outputs/eval/dpo_vs_sft_judged.jsonl \  
  --judge_model google/gemini-3.6-flash \  
  --batch_size 5 \  
  --sleep_seconds 5 \  
  --max_retries 3 \  
  --reuse_judgments
```

The judge should be instructed to compare responses based on helpfulness, correctness, instruction-following, concision when appropriate, formatting, and absence of hallucinations or unsafe content. For safety-focused evaluation sets, the prompt should also tell the judge to reward appropriate refusal, privacy protection, calibrated uncertainty, and safe redirection.

### Do Not Hammer the Judge Model

Large, fast request loops are more likely to fail or be denied. Prefer small batches, a 5-second delay, retries with logging, and resumable output files. This is slower but much more reliable.

## 14 Task 8: Final Report

Your final report should be concise and focused. It should include:

1. the GitHub repository link;
2. the model and datasets used;
3. the SFT, HH-RLHF DPO, UltraFeedback DPO, and LoRA-DPO training setup;
4. plots for the training runs, such as training loss, validation loss, reward accuracy, or reward margin where applicable;

5. DPO-vs-SFT evaluation results;
6. a short analysis explaining why DPO did or did not perform better than SFT;
7. any deviations from the guide configuration, including decoding settings such as generation penalties.

Keep the report direct. You do not need a long qualitative error analysis section, generation-throughput table, or broad quantitative analysis beyond the metrics needed to support your conclusion.

## 15 Expected Runtime

For the recommended 30k SFT, two 15k DPO runs, LoRA-DPO, and 300-prompt evaluation setup, a rough estimate on Kaggle T4x2 is:

| Stage                          | Estimated time |
|--------------------------------|----------------|
| SFT training                   | 1.5 hours      |
| Each from-scratch DPO training | 2.0–2.5 hours  |
| LoRA-DPO training              | 0.1–0.3 hours  |
| Generation                     | 0.5–1.0 hours  |
| Kaggle model judging           | 0.3–0.5 hours  |
| Total                          | 6.5–8.5 hours  |

### Runtime Planning

Budget an eight- to nine-hour Kaggle session to be safe if you run every track end to end. Save notebook versions and archive important outputs as soon as a stage completes.

## 16 Submission Checklist

Submit:

1. GitHub repository URL for your completed project.
2. A short final report as PDF.
3. Training plots for SFT and DPO runs.
4. Evaluation summary comparing DPO against SFT.

## 17 Academic Integrity and Collaboration

You may discuss concepts, debugging strategies, and experimental observations with classmates. Your implementation and report must be your own. If you use external references, cite them. Do not submit code copied from public DPO implementations as your own. You may consult documentation for PyTorch, Hugging Face Transformers, Kaggle, and the DPO paper.

## 18 Resources

### Useful Links

#### Hugging Face

[Transformers docs](#)  
[Qwen2.5-0.5B model card](#)

#### PEFT / LoRA

[PEFT docs](#)

#### Preference Optimization

[DPO paper](#)  
[TRL docs](#)  
[DPOTrainer guide](#)

#### Kaggle

[Notebooks docs](#)  
[Models docs](#)  
[GPU usage](#)

## References

1. Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C. D., and Finn, C. [Direct Preference Optimization: Your Language Model is Secretly a Reward Model](#). NeurIPS, 2023.
2. Ouyang, L. et al. [Training language models to follow instructions with human feedback](#). NeurIPS, 2022.
3. Li, X. et al. [AlpacaEval: An Automatic Evaluator of Instruction-following Models](#). 2023.
4. Bai, Y. et al. [Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback](#). 2022.
5. Hugging Face. [Transformers documentation](#), [TRL documentation](#), and [PEFT documentation](#).
6. Kaggle. [Notebooks documentation](#), [Models documentation](#), and [efficient GPU usage guide](#).